Assignment 4
(Oct. 5-19)

Problem 1:

The following LP is in SEF:

$$\begin{aligned} &\max \quad (1,-2,0,1,3)x \\ &\text{subject to} \\ &\qquad \begin{pmatrix} 1 & -1 & 2 & -1 & 0 \\ 2 & 0 & 1 & -1 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \\ &\qquad x \geq \mathbf{0} \end{aligned} \tag{1}$$

Find an equivalent LP in canonical form for:

(a) the basis $\{1,4\}$,

(b) the basis $\{3,5\}$.

In each case, state whether the basis (i.e. the corresponding basic feasible solution) is feasible.

Problem 2:

Consider the following parameterized linear program in standard equality form, where $\alpha \in \mathbb{R}$:

$$\begin{aligned} \max \quad & x_4 \\ \text{s.t.} \quad & -x_1 + x_3 = 1, \\ & x_2 + \alpha x_3 + x_4 = 0, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{P_α}$$

(a) Find a certificate of infeasibility of (P_α) for $\alpha = 1$.

(b) Find a certificate of unboundedness of (P_α) for $\alpha = -1$.

(c) Determine all values of α for which (P_α) is infeasible. Provide a certificate of infeasibility for each such α.

(d) Determine all values of α for which (P_α) is unbounded. Provide a certificate of unboundedness for each such α.

(e) For any remaining feasible values of α, find an optimal solution $\bar{x}$ and a certificate for the optimality of $\bar{x}$.

Problem 3:

In this exercise, you will work through a guided application of the simplex method to an LP. You do not need to have learned all the details of the algorithm in order to complete the exercise.

Consider the following LP

$$\begin{aligned} &\max \begin{pmatrix} 1 & 1 & 0 & 0 \end{pmatrix} x \\ &s.t. \begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &\quad x \geq \mathbf{0}. \end{aligned}$$

(a) Let $\bar{x}$ be the basic feasible solution with basis $\{3, 4\}$. What is $\bar{x}$?

(b) Explain why $\bar{x}$ is not an optimal solution.

(c) Let $x^{(\alpha)}$ be the solution to

$$\begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

such that $x_1^{(\alpha)} = \alpha$ and $x_2^{(\alpha)} = 0$. Determine the set $\mathcal{A} = \{\alpha : \ x^{(\alpha)} \text{ is a feasible solution}\}$.

(d) Let α^* be the maximum value of α in $\mathcal{A}$ (from part (c) you should see that the maximum exists). Show that $x^{(\alpha^*)}$ is a basic feasible solution whose objective value is greater than that for $\bar{x}$. What is the basis for $x^{(\alpha^*)}$?

(e) Redo part (c) but swapping the role of x_1 and x_2. Let $x^{(\alpha)}$ be the solution to

$$\begin{pmatrix} 2 & -1 & 1 & 0 \\ 1 & -2 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

such that $x_1^{(\alpha)} = 0$ and $x_2^{(\alpha)} = \alpha$. Determine the set $\mathcal{B} = \{\alpha : \ x^{(\alpha)} \text{ is a feasible solution}\}$. What does $\mathcal{B}$ imply about the three possible outcomes of this LP?

Problem 4:

Consider the following linear program in standard equality form:

$$\begin{aligned} \min \quad & x_4 \\ \text{s.t.} \quad & x_1 + x_2 + x_3 = 3, \\ & x_1 + 2x_2 + 3x_3 + x_4 = 6, \\ & x_1, x_2, x_3, x_4 \geq 0. \end{aligned} \tag{P}$$

In matrix form, the constraints are $Ax = b$, $x \geq 0$, where

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 3 \\ 6 \end{pmatrix}.$$

Let $\bar{x}$ be a feasible solution given by $(1\ 1\ 1\ 0)^T$. Let $J = \{1, 2, 3\}$ be the set of indices corresponding to the positive coordinates of $\bar{x}$.

(a) Find a nonzero vector $d \in \mathbb{R}^4$ such that

$$Ad = 0$$

and d is supported on J. That is, $d_j = 0$ for every $j \notin J$.

(b) For $\alpha \in \mathbb{R}$, define

$$x^{(\alpha)} = \bar{x} + \alpha d.$$

Show that $Ax^{(\alpha)} = b$ for every $\alpha \in \mathbb{R}$. Determine $\mathcal{A}$, the set of all values of α for which $x^{(\alpha)}$ is feasible.

(c) Choose α^* to be either $\max\{\alpha : \alpha \in \mathcal{A}\}$ or $\min\{\alpha : \alpha \in \mathcal{A}\}$, whichever exists. Compute the resulting feasible solution $x^* = x^{(\alpha^*)}$ and compare the number of its positive coordinates with that of $\bar{x}$.

(d) Prove that every feasible LP in SEF has a basic feasible solution by generalising the above arguments as follows. Consider LP: $\max\{c^T x : Ax = b, x \geq \mathbf{0}\}$. Let $\bar{x}$ be a feasible solution with minimum number of positive entries. Let J be the set of indices corresponding to the positive coordinates of $\bar{x}$. Following the guidance below to show that $\{A_j : j \in J\}$ is linearly independent.

(i) Suppose on the contrary that $\{A_j : j \in J\}$ is not linearly independent. Show that there exists a nonzero d supported on J such that $Ad = 0$.

(ii) Explain why without loss of generality we may assume that $d_j < 0$ for some $j \in J$.

(iii) For $\alpha \in \mathbb{R}$, define

$$x^{(\alpha)} = \bar{x} + \alpha d.$$

Show that $Ax^{(\alpha)} = b$ for every $\alpha \in \mathbb{R}$.

(iv) Let $\mathcal{A}$ be the set of all values of α for which $x^{(\alpha)}$ is feasible. Show that $\mathcal{A}$ has a maximum.

(v) Let α^* be the maximum of $\mathcal{A}$ and let $x^* = x^{(\alpha^*)}$. Show that the number of positive coordinates in x^* is smaller than $|J|$.

(vi) Conclude that $\{A_j : j \in J\}$ is linear independent. Explain why $\bar{x}$ is a basic feasible solution.

Solutions

Problem 1

For (a) we have

$$\begin{aligned} &\max \quad (0,-5,4,0,1)x - 5 \\ &\text{subject to} \\ &\qquad \begin{pmatrix} 1 & 1 & -1 & 0 & 1 \\ 0 & 2 & -3 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} -2 \\ -3 \end{pmatrix} \\ &\qquad x \geq \mathbf{0} \end{aligned} \tag{2}$$

The associated basic solution is $x = (-2, 0, 0, -3, 0)^T$, which is not feasible. For (b) we have

$$\begin{aligned} &\max \quad (-7/2, -7/2, 0, 5/2, 0)x - 9/2 \\ &\text{subject to} \\ &\qquad \begin{pmatrix} 1/2 & -1/2 & 1 & -1/2 & 0 \\ 3/2 & 1/2 & 0 & -1/2 & 1 \end{pmatrix} x = \begin{pmatrix} 1/2 \\ -3/2 \end{pmatrix} \\ &\qquad x \geq \mathbf{0} \end{aligned} \tag{3}$$

The associated basic solution is $x = (0, 0, 1/2, 0, -3/2)^T$, which is not feasible.

We arrive on these formulas via row operations. For (a), start from (1) and subtract twice the first row from the second to get

$$\begin{aligned} &\max \quad (1,-2,0,1,3)x \\ &\text{subject to} \\ &\qquad \begin{pmatrix} 1 & -1 & 2 & -1 & 0 \\ 0 & 2 & -3 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 1 \\ -3 \end{pmatrix} \\ &\qquad x \geq \mathbf{0} \end{aligned}$$

We now add the second row to the first one to get

$$\begin{aligned} &\max \quad (1,-2,0,1,3)x \\ &\text{subject to} \\ &\qquad \begin{pmatrix} 1 & 1 & -1 & 0 & 1 \\ 0 & 2 & -3 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} -2 \\ -3 \end{pmatrix} \\ &\qquad x \geq \mathbf{0} \end{aligned}$$

We now need to fix the objective value. You can simply multiply the inequalities on the left by the row vector $c_B = (1, 1)^T$ and use it to simplify the objective value. Equivalently, you can subtract the first row from the objective value while adding the RHS, which amounts to substituting the expression for x_1. Doing so we get that

$$\begin{aligned} &\max \quad (0,-3,1,1,2)x - 2 \\ &\text{subject to} \\ &\qquad \begin{pmatrix} 1 & 1 & -1 & 0 & 1 \\ 0 & 2 & -3 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} -2 \\ -3 \end{pmatrix} \\ &\qquad x \geq \mathbf{0} \end{aligned}$$

We now do the same for the second row, this substituting out x_4:

$$\begin{aligned} \max \quad & (0,-5,4,0,1)x - 5 \\ \text{subject to} \quad & \\ & \begin{pmatrix} 1 & 1 & -1 & 0 & 1 \\ 0 & 2 & -3 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} -2 \\ -3 \end{pmatrix} \\ & x \geq \mathbf{0} \end{aligned}$$

For (b), set $B = \{3, 5\}$. We will compute the inverse of

$$A_B = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}.$$

We can accomplish this via row operations again. By first scaling the first row by $\frac{1}{2}$ and then subtracting the first row from the second row, we get

$$\begin{pmatrix} 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 1/2 & 0 \\ 0 & 1 & -1/2 & 1 \end{pmatrix}$$

Hence

$$A_B^{-1} = \begin{pmatrix} 1/2 & 0 \\ -1/2 & 1 \end{pmatrix}.$$

We can thus simply compute $A_B^{-1}A$ and $A_B^{-1}b$:

$$A_B^{-1}A = \begin{pmatrix} 1/2 & 0 \\ -1/2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 & -1 & 0 \\ 2 & 0 & 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & -1/2 & 1 & -1/2 & 0 \\ 3/2 & 1/2 & 0 & -1/2 & 1 \end{pmatrix},$$

$$A_B^{-1}b = \begin{pmatrix} 1/2 & 0 \\ -1/2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -3/2 \end{pmatrix}.$$

Finally, we simply compute $c_B^T A_B^{-1} A$ and $c_B^T A_B^{-1} b$ to now how to modify our objective value.

We have that

$$\begin{aligned} \begin{pmatrix} 9/2 & 3/2 & 0 & -3/2 & 3 \end{pmatrix} x &= \begin{pmatrix} 0 & 3 \end{pmatrix} \begin{pmatrix} 1/2 & -1/2 & 1 & -1/2 & 0 \\ 3/2 & 1/2 & 0 & -1/2 & 1 \end{pmatrix} x \\ &= c_B^T A_B^{-1} A x \\ &= c_B^T A_B^{-1} b \\ &= \begin{pmatrix} 0 & 3 \end{pmatrix} \begin{pmatrix} 1/2 \\ -3/2 \end{pmatrix} \\ &= -9/2. \end{aligned}$$

Hence

$$\begin{aligned} \begin{pmatrix} 1 & -2 & 0 & 1 & 3 \end{pmatrix} x &= \Big(\begin{pmatrix} 1 & -2 & 0 & 1 & 3 \end{pmatrix} - \begin{pmatrix} 9/2 & 3/2 & 0 & -3/2 & 3 \end{pmatrix} \Big) x - 9/2 \\ &= \begin{pmatrix} -7/2 & -7/2 & 0 & 5/2 & 0 \end{pmatrix} x - 9/2. \end{aligned}$$

We thus have arrived at the LP in (3).

Problem 2

We write the equations in (P_α) by $A_\alpha x = b$ where

$$A_\alpha = \begin{pmatrix} -1 & 0 & 1 & 0 \\ 0 & 1 & \alpha & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Let $c = (0\ 0\ 0\ 1)^T$. Thus, the objective function is $c^T x$.

(a) With $\alpha = 1$, we find that $y = (-1\ 1)^T$ is a certificate for the infeasibility since

$$A_1^T y = (1\ 1\ 0\ 1)^T \geq \mathbf{0}, \quad b^T y = -1 < 0.$$

(b) With $\alpha = -1$, we find that $(\bar{x}, d)$ given by

$$\bar{x} = (0\ 0\ 1\ 1)^T, \quad d = (1\ 0\ 1\ 1)^T$$

is a certificate for unboundedness since $\bar{x}$ is a feasible solution and

$$A_{-1} d = 0,\ d \geq \mathbf{0},\ c^T d = 1 > 0.$$

(c,d,e): The first equation gives

$$x_3 = 1 + x_1.$$

Since $x_1 \geq 0$, every feasible solution must satisfy

$$x_3 \geq 1.$$

The second equation can be written as

$$x_2 + x_4 = -\alpha x_3.$$

Because $x_2, x_4 \geq 0$, the left-hand side is nonnegative. It immediately implies that if $\alpha > 0$ then (P_α) is infeasible. For every $\alpha > 0$, $y = (-1\ 1/\alpha)^T$ is a certificate for infeasibility, since $A_\alpha^T y = (1\ 1/\alpha\ 0\ 1/\alpha) \geq \mathbf{0}$ and $b^T y = -1 < 0$.

On the other hand, for every $\alpha \leq 0$, $\bar{x} = (0\ 0\ 1\ -\alpha)^T$ is a feasible solution and hence (P_α) is feasible. Suppose $\alpha \leq 0$. To see whether (P_α) is unbounded, we find that the null space of A_α is the span of $(1\ 0\ 1\ -\alpha)^T$ and $(0\ 1\ 0\ -1)^T$.

Let $d = (1\ 0\ 1\ -\alpha)^T$. Then it satisfies that $d \geq \mathbf{0}$ and $A_\alpha d = 0$. Moreover, it satisfies $c^T d > 1$ if $\alpha < 0$.

Hence, for all $\alpha < 0$, P_α is unbounded with a certificate of unboundedness given by $(\bar{x}, d)$ where

$$\bar{x} = (0\ 0\ 1\ -\alpha)^T, \quad d = (1\ 0\ 1\ -\alpha)^T.$$

The only remaining value for α is $\alpha = 0$. In this case, every feasible solution x must satisfy $x_3 = 1 + x_1$ and $x_4 = -x_2$. But $x \geq \mathbf{0}$ which implies that $x_2 = x_4 = 0$. Hence, for every feasible x, its objective value is equal to zero. Thus, every feasible solution is optimal. For instance $\bar{x} = (0\ 0\ 1\ 0)^T$ is an optimal solution, and $y = (0\ 1)^T$ is a certificate for its optimality, since $c - A_0^T y = (0\ -1\ 0\ 0)^T \leq \mathbf{0}$ and $c^T \bar{x} = b^T y = 0$.

Notice that when $\alpha = 0$, the feasible region is itself unbounded because x_1 and $x_3 = 1 + x_1$ may grow without bound. Nevertheless, the linear program is not unbounded. An unbounded feasible region does not necessarily imply that an LP is unbounded.

Problem 3

(a) $\bar{x} = (0\ 0\ 1\ 1)^T$.

(b) The objective value for $\bar{x}$ is 0. However, $x' = (0\ 1\ 2\ 3)^T$ is a feasible solution with objective value 1 which is greater than 0. Hence $\bar{x}$ is not an optimal solution.

(c) Setting $x_1 = \alpha$ and $x_2 = 0$, the linear system has a unique solution $x_3 = 1 - 2\alpha$ and $x_4 = 1 - \alpha$. The feasibility of $x^{(\alpha)}$ requires $x_1, x_3, x_4 \geq 0$ which is equivalent to $0 \leq \alpha \leq 1/2$. Hence, $\mathcal{A} = [0, 1/2]$.

(d) By $\mathcal{A} = [0, 1/2]$ we have $\alpha^* = 1/2$ and $x^{(\alpha^*)} = (1/2\ 0\ 0\ 1/2)^T$. It is a basic feasible solution with basis $\{1, 4\}$ and its objective value is equal to $1/2 > 0$.

(e) Setting $x_1 = 0$ and $x_2 = \alpha$, the linear system has a unique solution $x_3 = 1+\alpha$ and $x_4 = 1+2\alpha$. The feasibility of $x^{(\alpha)}$ requires $x_2, x_3, x_4 \geq 0$ which is equivalent to $\alpha \geq 0$. Hence, $\mathcal{B} = [0, +\infty)$. This implies that the LP is unbounded since $\lim_{\alpha\to\infty}(1\ 1\ 0\ 0)^T x^{(\alpha)} = \lim_{\alpha\to\infty} \alpha = \infty$.

Problem 4

(a) We seek a nonzero vector of the form

$$d = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \\ 0 \end{pmatrix}$$

satisfying $Ad = 0$. This gives

$$\begin{aligned} d_1 + d_2 + d_3 &= 0, \\ d_1 + 2d_2 + 3d_3 &= 0. \end{aligned}$$

Solving this linear system we find that $d = t(1\ -2\ 1\ 0)^T$ is a solution for every $t \in \mathbb{R}$. We may choose

$$d = \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix}.$$

(b) For every $\alpha \in \mathbb{R}$,

$$Ax^{(\alpha)} = A(\bar{x} + \alpha d) = A\bar{x} + \alpha Ad = b.$$

Moreover,

$$x^{(\alpha)} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1+\alpha \\ 1-2\alpha \\ 1+\alpha \\ 0 \end{pmatrix}.$$

Feasiblity of $x^{(\alpha)}$ requires

$$1 + \alpha \geq 0 \qquad \text{and} \qquad 1 - 2\alpha \geq 0.$$

Hence

$$\mathcal{A} = [-1, 1/2].$$

(c) Both the maximum and the minimum of $\mathcal{A}$ exist and we can choose either. Let us choose

$$\alpha^\star = \frac{1}{2}.$$

Substituting this value gives

$$x^* = \begin{pmatrix} 1 + \frac{1}{2} \\ 1 - 2\left(\frac{1}{2}\right) \\ 1 + \frac{1}{2} \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \\ 0 \\ \frac{3}{2} \\ 0 \end{pmatrix}.$$

Compared to $\bar{x}$, x^* has less positive coordinates.

(d) Let

$$J = \{j : \bar{x}_j > 0\}.$$

We show that the columns $\{A_j : j \in J\}$ are linearly independent.

(i) Suppose, for contradiction, that

$$\{A_j : j \in J\}$$

is linearly dependent. Then there exist scalars d_j, not all zero, such that

$$\sum_{j \in J} d_j A_j = 0.$$

Define $d \in \mathbb{R}^n$ by

$$d_j = \begin{cases} d_j, & j \in J, \\ 0, & j \notin J. \end{cases}$$

Thus d is nonzero and supported on J. Moreover,

$$Ad = \sum_{j=1}^{n} d_j A_j = \sum_{j \in J} d_j A_j = 0.$$

(ii) Since $d \neq 0$, there is at least one $j \in J$ for which $d_j \neq 0$. If $d_j < 0$ no change is needed. Otherwise, replace d by $-d$. The new vector is still nonzero, is still supported on J, and satisfies

$$A(-d) = -Ad = 0.$$

(iii) Since $\bar{x}$ is feasible, $A\bar{x} = b$. Since $Ad = 0$, we have

$$Ax^{(\alpha)} = A(\bar{x} + \alpha d) = A\bar{x} + \alpha Ad = b + \alpha \mathbf{0} = b.$$

(iv) Let

$$\mathcal{A} = \{\alpha \in \mathbb{R} : x^{(\alpha)} \geq 0\}.$$

Because $x^{(0)} = \bar{x}$ is feasible, we have $0 \in \mathcal{A}$, so $\mathcal{A}$ is nonempty.

Define

$$J^- = \{j \in J : d_j < 0\}.$$

By part (ii), J^- is nonempty. For any $j \in J^-$, the nonnegativity condition on $x_j^{(\alpha)}$ requires

$$\bar{x}_j + \alpha d_j \geq 0.$$

Since $d_j < 0$, this is equivalent to

$$\alpha \leq \frac{\bar{x}_j}{-d_j}.$$

Consequently, every $\alpha \in \mathcal{A}$ satisfies

$$\alpha \leq \min_{j \in J^-} \frac{\bar{x}_j}{-d_j}.$$

Define

$$\alpha^* = \min_{j \in J^-} \frac{\bar{x}_j}{-d_j}.$$

It suffices to verify that $\alpha^* \in \mathcal{A}$, which then implies that $\alpha^* = \max\{\alpha : \alpha \in \mathcal{A}\}$.

Because $\bar{x}_j > 0$ for every $j \in J$ and $-d_j > 0$ for every $j \in J^-$, we have

$$\alpha^* > 0.$$

For any $j \in J^-$, the definition of α^* gives

$$\alpha^* \leq \frac{\bar{x}_j}{-d_j},$$

and hence

$$\bar{x}_j + \alpha^* d_j \geq 0.$$

For any $j \in J \setminus J^-$, $d_j \geq 0$, and so

$$\bar{x}_j + \alpha^* d_j \geq \bar{x}_j > 0.$$

Finally, if $j \notin J$, then $\bar{x}_j = 0$ and $d_j = 0$, because d is supported on J. Therefore,

$$\bar{x}_j + \alpha^* d_j = 0.$$

It follows that

$$x^{(\alpha^*)} \geq 0.$$

This verifies that $\max\{\alpha \in \mathcal{A}\}$ exists, and is given by α^*.

(v) By the definition of α^*, there exists some $k \in J^-$ such that

$$\alpha^* = \frac{\bar{x}_k}{-d_k}.$$

For this index,

$$x_k^* = \bar{x}_k + \alpha^* d_k = \bar{x}_k + \frac{\bar{x}_k}{-d_k} d_k = 0.$$

Hence at least one coordinate that was positive in $\bar{x}$ has become zero in x^*.

Also, no new positive coordinates are introduced outside J. For every $j \notin J$,

$$x_j^* = \bar{x}_j + \alpha^* d_j = 0,$$

since $\bar{x}_j = 0$ and $d_j = 0$. Therefore, the number of positive coordinates in x^* is smaller than $|J|$.

(vi) We have shown that x^* is feasible and has fewer positive coordinates than $\bar{x}$. This contradicts the choice of $\bar{x}$ as a feasible solution with the minimum possible number of positive coordinates. Therefore, $\{A_j : j \in J\}$ must be linearly independent. By A1-Problem 4, J can be extended to a basis B of A. Moreover, $\bar{x}_j = 0$ for all $j \notin B$. Hence $\bar{x}$ is a basic feasible solution.